ALICE L WALTON SCHOOL OF MEDICINE

Pressure Volume (PV) Loops

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Adapted from: Pressure-Volume Loops

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Analyze and interpret a ventricular pressure-volume loop.
2. Relate the ventricular pressure-volume loop to the events of the cardiac cycle.
3. Determine stroke volume, ejection fraction, and stroke work from a pressure-volume loop.
4. Describe how pressure-volume loops are affected by preload, afterload, and autonomic nervous system activity.

CASE CONNECTION

JR and her friends decide to see a horror movie at a local theater. Before the movie begins, JR talks to her friends; she's relaxed and doesn't notice her own heart rate. Once the plot starts to unfold, JR becomes terrified, and she feels her heart rate jump. At the end of the movie, although she is calm again, she tells her friends, "I don't think I will be able to sleep tonight!"

If JR asked you about her body's reaction, how would you explain her hemodynamic changes during the movie? Consider your answers as you read, and we'll revisit JR at the end of the brick.

GO TO CONCLUSION ↓

What Are Pressure–Volume Loops (LO1)?

The heart is a pump, but how can we calculate how much work it does and how efficiently it is pumping, or even demonstrate different valvular disease states? As the heart moves through each cycle of contraction (systole) and relaxation (diastole) to create a heartbeat, pressure and volume changes occur in the ventricles. In other bricks, you have learned about the pressures and volumes in the cardiac chambers during the various parts of the cardiac cycle. It's now time to put together this knowledge in one diagram.

Visually, these chang pressure-volume (PV) loops: pressure is indicated on the y-axis, volume is indicated on the x-axis, and as the points are plotted through one complete cardiac cycle, they create a loop (Figure 1).

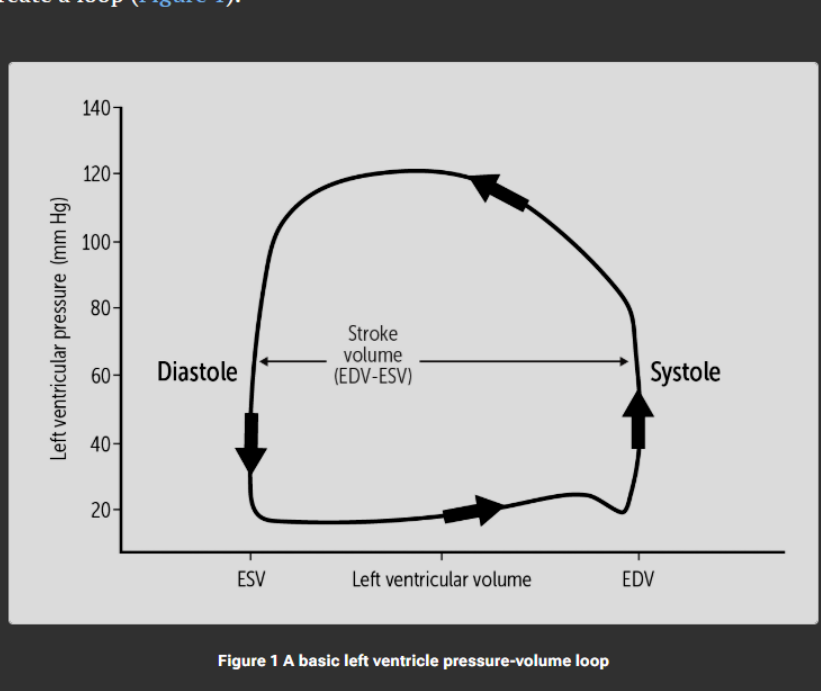


Figure 1 A basic left ventricle pressure-volume loop

Measurements taken from the pressure-volume loops can then be used to calculate other parameters, including **stroke volume** and **ejection fraction**, both measures of cardiovascular performance.

Review of the Card

Before describing the PV loop in detail, let's show the cardiac cycle were the pressure and volume of the left ventricle (LV) change over time. Both of these depictions show similar changes, but from different perspectives as the heart goes through the seven phases of the cardiac cycle (Figure 2).

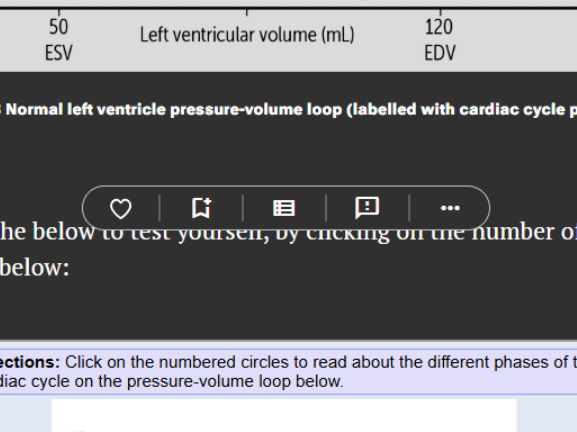


Figure 2 Pressure and volume changes in the cardiac cycle

Normal PV Loop related to the Cardiac Cycle (LO2)

Now let's see how th ure-volume loop. It can be a bit confusing at first, since *now the x-axis does not represent time*, unlike the cardiac cycle in Figure 2.

The seven phases of the cardiac cycle are shown on a pressure-volume loop, where left ventricular volume is on the x-axis and left ventricular pressure is on the y-axis (Figure 3). (The PV loops for the LV are commonly shown because they are more rectangular, have higher pressures, and have more defined isovolumetric phases). Note that while a pressure-volume loop can be drawn for the right ventricle, it's more typical to see these for the left ventricle.

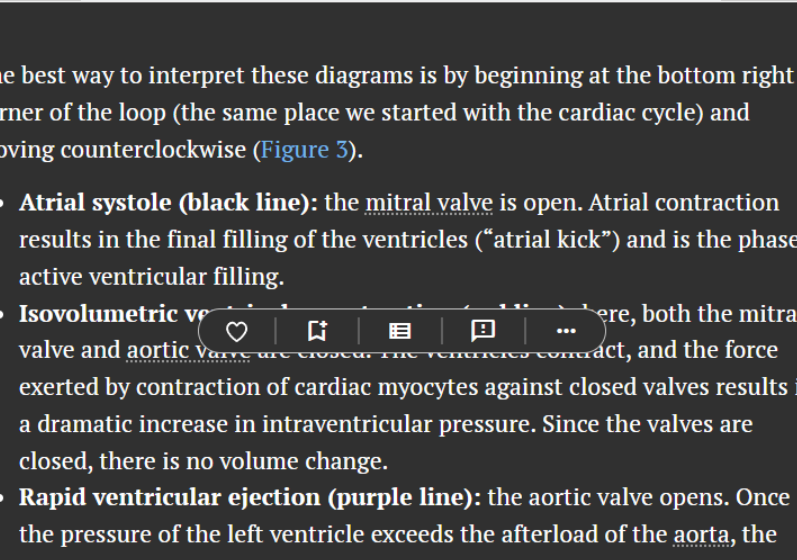
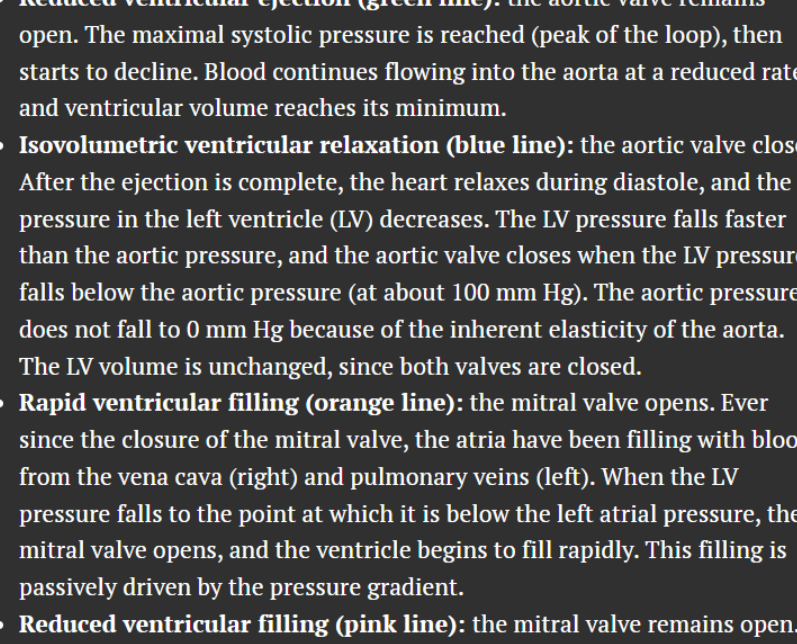


Figure 3 Normal left ventricle pressure-volume loop (labelled with cardiac cycle phases)

You can use the below to test yourself, by clicking on the number of each phase in the image below:


Directions: Click on the numbered circles to read about the different phases of the cardiac cycle on the pressure-volume loop below.

The best way to interpret these diagrams is by beginning at the bottom right corner of the loop (the same place we started with the cardiac cycle) and moving counterclockwise (Figure 3).

- **Atrial systole (black line):** the mitral valve is open. Atrial contraction results in the final filling of the ventricles ("atrial kick") and is the phase of active ventricular filling.
- **Isovolumetric ve** ere, both the mitral valve and aortic va contract, and the force exerted by contraction of cardiac myocytes against closed valves results in a dramatic increase in intraventricular pressure. Since the valves are closed, there is no volume change.
- **Rapid ventricular ejection (purple line):** the aortic valve opens. Once the pressure of the left ventricle exceeds the afterload of the aorta, the aortic valve opens and blood is rapidly ejected into the aorta, decreasing left ventricular volume (observed traveling counterclockwise on the graph).
- **Reduced ventricular ejection (green line):** the aortic valve remains open. The maximal systolic pressure is reached (peak of the loop), then starts to decline. Blood continues flowing into the aorta at a reduced rate, and ventricular volume reaches its minimum.
- **Isovolumetric ventricular relaxation (blue line):** the aortic valve closes. After the ejection is complete, the heart relaxes during diastole, and the pressure in the left ventricle (LV) decreases. The LV pressure falls faster than the aortic pressure, and the aortic valve closes when the LV pressure falls below the aortic pressure (at about 100 mm Hg). The aortic pressure does not fall to 0 mm Hg because of the inherent elasticity of the aorta. The LV volume is unchanged, since both valves are closed.
- **Rapid ventricular filling (orange line):** the mitral valve opens. Ever since the closure of the mitral valve, the atria have been filling with blood from the vena cava (right) and pulmonary veins (left). When the LV pressure falls to the point at which it is below the left atrial pressure, the mitral valve opens, and the ventricle begins to fill rapidly. This filling is passively driven by the pressure gradient.
- **Reduced ventricular filling (pink line):** the mitral valve remains open. Passive filling of alve, but the difference in pres d LV is less, so filling slows.

What phases occur when the aortic and mitral valves are closed?

How Do We Determine Blood Pressure, Stroke Volume, Stroke Work, and Ejection Fraction From the Loop (LO3)?

We'll now use the loops to demonstrate some key concepts—blood pressure, stroke volume, and stroke work.

Arterial Blood Pressure

The systolic and diastolic blood pressures are usually measured with a blood pressure cuff; the pressures are normally 120 mm Hg (systolic) and 80 mm Hg (diastolic). How can re from a pressure-volume loop?

Figure 4 shows that **diastolic BP** is estimated by the pressure when the aortic valve is closed, and **systolic BP** is estimated at the peak of the loop. The peak of the loop is not only the maximal LV systolic pressure, but because the aortic valve is open, it is also the maximal systolic arterial blood pressure.

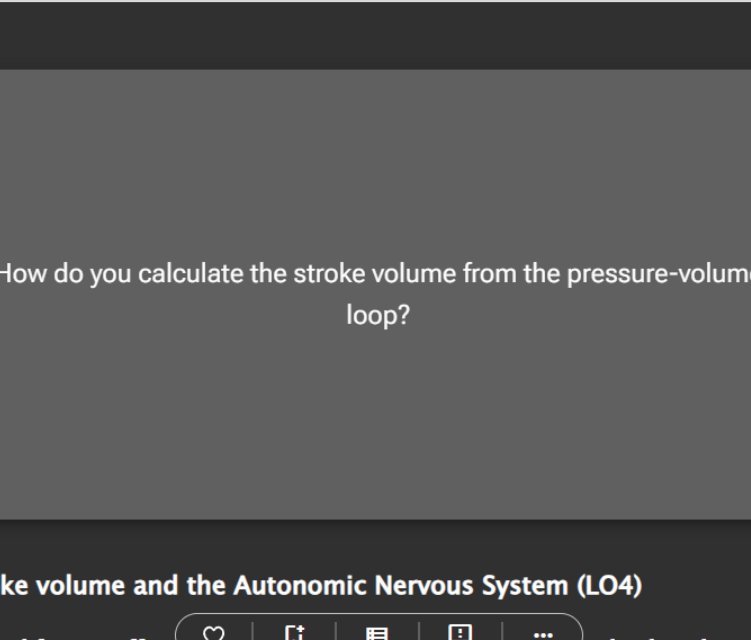


Figure 4. Blood pressures, stroke volume (arrow) and stroke work (blue shading) related to the PV loop

Stroke Volume (SV)

Stroke volume is th ejected out of the left ventricle into the aorta to supply the rest of the body (systemic circulation). SV can be calculated as the difference between **end-diastolic volume (EDV)** and **end-systolic volumes (ESV)** and is seen in the horizontal displacement (width) of the loop (Figure 4). Recall that EDV is related to preload, and thus, preload directly affects SV.

SV = EDV – ESV

How do you calculate the stroke volume from the pressure-volume loop?

Stroke volume and the Autonomic Nervous System (LO4)

Several factors affect preload, and afterload. However, the **autonomic nervous system (ANS)** also affects the SV. An increase in heart rate reduces the time the heart can spend in diastolic filling! As a result, the EDV, preload, and consequently the stroke volume decrease.

Stroke Work

In the cardiac system, **stroke work** can be defined as the force exerted by the left ventricle multiplied by the volume ejected from the ventricle. This means that stroke work is equivalent to the **area within the pressure-volume loop curve**. Stroke work is represented by the green shaded area in the center of the loop (Figure 4). The bigger the area, the more the work. So a heart that has to pump against higher blood pressure, meaning the aortic valve opens later, will have a taller loop and greater area, meaning more work.

CLINICAL CORRELATION

Systolic heart failure occurs when the amount of blood ejected during systole falls, and the end-systolic pressure increases. This decreases the stroke volume (less blood ejected into the aorta), sometimes leading to fatigue and dizziness.

Ejection Fraction

Ejection fraction (EF) is a measure of how effectively the heart pumps blood with each beat. It represents the percentage of blood in the left ventricle that is ejected during systol heart is functioning as a pump, and getting blood into the circulation. EF is calculated using the formula:

$$EF = \left(\frac{\text{Stroke Volume}}{\text{End-Diastolic Volume}}\right) \times 100\%$$

A normal ejection fraction is typically **greater than 55%**, meaning that more than half of the blood in the ventricle is ejected with each beat. If EF is lower than normal, it suggests that the heart is not pumping efficiently.

On a **pressure-volume loop**, a stronger contraction, results in a lower **end-systolic volume (ESV)**, or more blood is pushed out of the heart, and leads to a higher ejection fraction. In contrast, if the heart is not contracting as strongly, more blood remains in the ventricle after systole, leading to a lower EF. This concept is useful in understanding how different conditions affect heart function, such as changes in preload, afterload, and contractility, and is used to differentiate between reduced and preserved ejection fraction heart failure (e.g., Heart Failure with Reduced Ejection Fraction; HFrEF), and Heart Failure with Preserved Ejection Fraction (HFpEF), which will be discussed later on.

How Do Preload, Afterload, and Contractility Affect the Loop (LO4)?

Three important variab ure-volume loop are preload, afterload, an minants of stroke volume that impact cardiac output.

Preload effects on the PV loop

Preload is the amount of stretch in LV myocytes as a result of the EDV. The effects of increased and decreased preload in the left ventricle are shown in Figure 5, with the different preloads depicted as dots on the curve.

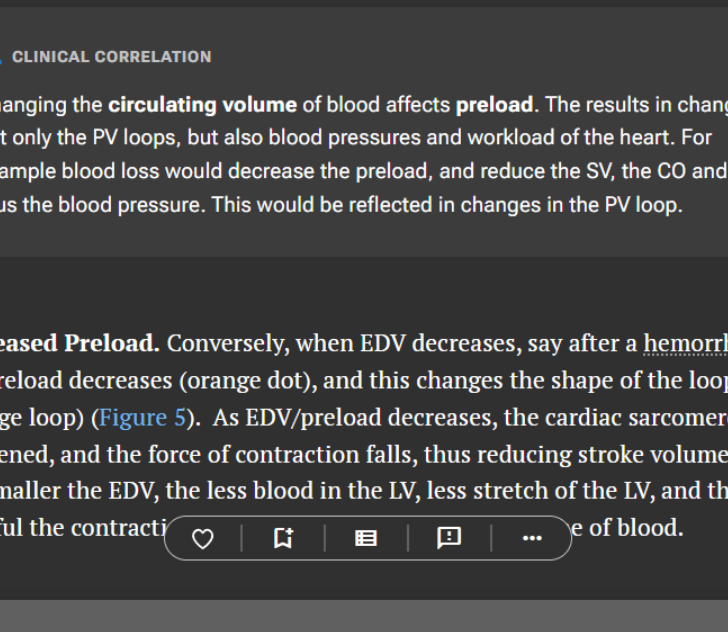


Figure 5 Effects of preload on the PV loop- altered EDV and SV

Increased Preload. a blood transfusion, preload increases (green dot), which changes the shape of the pressure–volume loop (green loop) (Figure 5). The dotted line represents the **end-systolic pressure–volume relationship (ESPVR)**. Recall the Frank–Starling law: increasing preload increases the developed force and shortening of contraction, but these changes occur **along the same ESPVR**, because the intrinsic contractile state of the myocardium (inotropy) has not changed. With greater preload, sarcomeres are more stretched at the onset of contraction, allowing greater shortening and pressure development. As a result, stroke volume increases, reflected by the wider loop, even though the ESPVR slope does not shift. In contrast, changes in contractility (inotropy) shift the ESPVR upward and to the left.

CLINICAL CORRELATION

Changing the **circulating volume** of blood affects **preload**. The results in changes not only the PV loops, but also blood pressures and workload of the heart. For example blood loss would decrease the preload, and reduce the SV, the CO and thus the blood pressure. This would be reflected in changes in the PV loop.

Decreased Preload. Conversely, when EDV decreases, say after a hemorrhage, the preload decreases (orange dot), and this changes the shape of the loop (orange loop) (Figure 5). As EDV/preload decreases, the cardiac sarcomeres are shortened, and the force of contraction falls, thus reducing stroke volume. So, the smaller the EDV, the less blood in the LV, less stretch of the LV, and the less forceful the contracti e of blood.

What effect does hemorrhage have on stroke volume?

What effect does an increase in preload have on the pressure-volume loop?

Afterload effects on the PV loop

Afterload describes the resistance to ejection from the left ventricle. This is usually the blood pressure in the aorta that the LV "sees" when the aortic valve opens. The effects of changing afterload are shown in Figure 6.

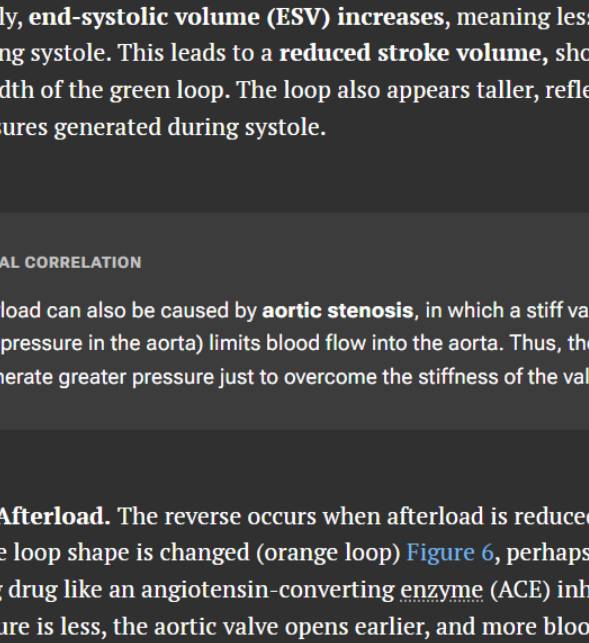


Figure 6 Effects of afterload on the PV loop- altered ESV, and SV

Increased Afterload. The green loop in Figure 6 shows how high afterload (different dots) compr that the end-diastolic volume (EDV in preload (x-axis). Contractility is also unchanged, so the ESPVR remains the same.

With higher afterload, the left ventricle must generate a **higher pressure before the aortic valve can open.** As a result, the aortic valve opens at a higher pressure (≈120 mm Hg, green dot) compared with the normal loop (≈80 mm Hg). Once ejection begins, the ventricle ejects blood (top horizontal line) until it reaches the ESPVR. However, because it is pumping against a higher pressure, myocardial fiber shortening is reduced, so the ventricle reaches the ESPVR at a larger end-systolic volume.

Consequently, **end-systolic volume (ESV) increases**, meaning less blood is ejected during systole. This leads to a **reduced stroke volume,** shown by the narrower width of the green loop. The loop also appears taller, reflecting the higher pressures generated during systole.

CLINICAL CORRELATION

High afterload can also be caused by **aortic stenosis**, in which a stiff valve (rather than high pressure in the aorta) limits blood flow into the aorta. Thus, the heart has to generate greater pressure just to overcome the stiffness of the valve.

Decreased Afterload. The reverse occurs when afterload is reduced (orange dot), and the loop shape is changed (orange loop) Figure 6, perhaps by taking a vasodilating drug like an angiotensin-converting enzyme (ACE) inhibitor. The aortic pressure is less, the aortic valve opens earlier, and more blood is ejected. This is why such drug

What effect does arterial vasodilation have on stroke volume?

Contractility effects on the PV loop

Contractility (also called inotropy) is the contractile force generated by myocytes independent of preload and afterload. It is typically due to changes in the intracellular calcium available to stimulate cardiomyocyte contraction, which is increased by sympathetic stimulation of the heart (beta-1 adrenergic receptor).

How does contractility look on the loop? You may have noticed the diagonal hatched line in the previous preload and afterload diagrams and in Figure 7. The dotted line is called the **end-systolic pressure-volume relationship (ESPVR)**—quite a mouthful. But the name makes sense, since the line represents the end-systolic pressures for a cardiac cycle at a variety of volumes (in the curves above, fterload, Figure 6, results in a similar inotropic response). The **slope** of the line represents the contractility. How the loops change with changes in contractility is seen in Figure 7.

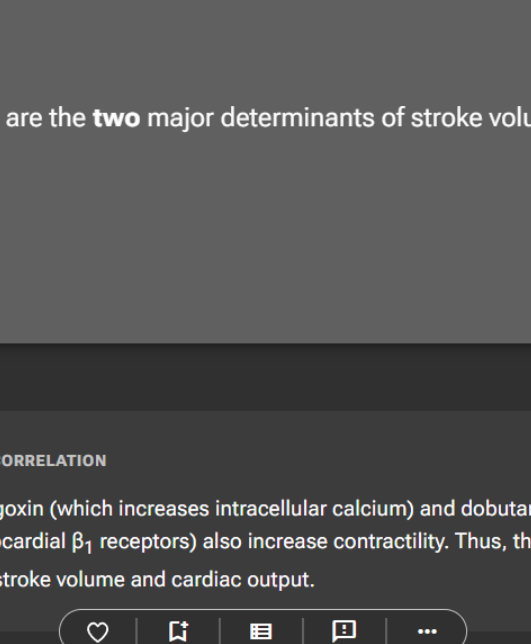


Figure 7 Effects of contractility on the PV loop- altered ESV, and SV

Increased Contractility. With increased contractility, we see a couple of things:

- ESPVR line **slope** becomes steeper: changes in cardiac ejection become more dramatic ar stolic volume.
- Most importantly, **the end-systolic volume.** Since the EDV is unchanged (preload unchanged), and ESV = EDV – SV, we conclude that there will be an increase in stroke volume (and cardiac output), just what we would expect with a more contractile heart.

Contractility commonly increases with increased sympathetic nerve input, as occurs during exercise.

What are the **two** major determinants of stroke volume?

CLINICAL CORRELATION

Drugs like digoxin (which increases intracellular calcium) and dobutamine (which activates myocardial β_1 receptors) also increase contractility. Thus, these drugs increase the stroke volume and cardiac output.

Decreased Contractility. The opposite is seen for decreased contractility Figure 7, as might occur after a myocardial infarction (MI) leading to systolic heart failure due to injured myocardium. Here, the end-systolic volume increases, the stroke volume decreases, and the cardiac output decreases.

What effect does myocardial infarction have on stroke volume?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to JR, how would you explain her hemodynamic changes during the movie?

JR's sympathetic nervous system is in full gear during the movie, causing her heart rate and her blood p ate reduces the time her heart can spend in diastolic filling, so EDV and preload decrease. As a result, stroke volume also decreases. The increase in blood pressure increases afterload. When JR finally calms down after the movie, her heart rate, blood pressure, EDV, preload, SV, and afterload promptly return to normal. JR decides that the next time she goes to a movie, she will choose a comedy.

Summary

- **Pressure-volume (PV) loops** show the relationship between ventricular pressure and volume, and is used to visualize cardiac performance.
- The events of the cardiac cycle can be shown on the PV loop and cardiac diagrams, but they differ in that no time is represented in the PV loop.
- **Stroke volume (SV)** is how much blood is ejected from the heart in one beat. It is calculated as SV = EDV – ESV or SV = CO/HR.
- **Stroke work** is the area enclosed by the pressure-volume curve.
- The **ejection fraction (EF)** is the ratio of blood volume pumped out of the ventricle; EF = SV/EDV. A normal EF is typically above 55%; lower EF implies diminished pump efficiency and is characteristic of systolic heart failure.
- Increasing **preload** increases the volume of blood in the heart at the end of systole, resulting in a higher EDV. Because the ventricle has more blood at the start of systole, stroke volume increases. The PV loop widens.
- Increasing **afterload** (e.g., from high arterial pressure or aortic stenosis) causes the ventricle to open the aortic valve at a higher pressure, with less blood ejected and a smaller SV. The PV loop is taller and narrower.
- Increased **contractility** increases the end-systolic pressure-volume relationship (ESI ower, and stroke volume increased. The PV loop widens.

Review Questions

1. Which of the following equations calculates the left ventricular stroke volume correctly?

A. ○ Volume at the beginning of atrial systole minus volume at the beginning of isovolumetric ventricular relaxation
B. ○ Volume at the beginning of isovolumetric ventricular contraction minus volume at the beginning of reduced ventricular filling
C. ● Volume at the end of isovolumetric ventricular contraction minus volume at the beginning of rapid ventricular filling
D. ○ Volume at the end of rapid ventricular filling minus volume at the end of rapid ventricular ejection
E. ○ Volume at the end of reduced ventricular filling minus volume at the end of reduced ventricular ejection

Hide Explanation

The correct answer is volume at the end of isovolumetric ventricular contraction minus volume at the beginning of rapid ventricular filling (C). The stroke volume (SV) is calculated by subtracting the end-systolic volume (ESV) from the end-diastolic volume (EDV): SV = EDV – ESV. Therefore, the left ventricular (LV) volume would be equal to the EDV during isovolumetric ventricular contraction (no volume change) and at the moments that immediately precede and follow that phase (ie, end of atrial systole and beginning of rapid ventricular ejection, respectively). Similarly, LV volume would be equ ular ejection, during the isovolumetric ven rapid ventricular filling. The LV volume at the beginning of atrial systole (as opposed to the end of atrial systole) is not equal to the EDV (A). The rapid ventricular ejection phase is immediately followed by the reduced ventricular ejection phase, therefore option (D) is not correct as the LV volume has not reached the ESV yet. The LV volume at the beginning of reduced ventricular filling (as opposed to the beginning of rapid ventricular filling) is not equal to the ESV (B). The volume at the end of reduced ventricular filling (as opposed to the end of atrial systole) is not equal to the EDV (E).

2. How does elevated systemic blood pressure affect the pressure-volume loop?

A. ○ Decrease in end-diastolic volume
B. ○ Decrease in end-systolic volume
C. ○ Increase in end-diastolic volume
D. ● Increase in end-systolic volume
E. ○ Increase in stroke volume

Hide Explanation

The correct answer is increase in end-systolic volume (D). Elevated blood pressure increases the afterload, restricting the extent by and the velocity at which the individual myocardial fibers contract. As a result, the volume of blood that is ejected by the heart on each cardiac cycle is decreased, leaving more blood behind at the end of each contract e end-diastolic volume (A, C), stroke volume, which is calculated by subtracting the end-systolic volume from the end-diastolic volume, would decrease (E).

3. Which of the following describes the effect of an intravenous infusion of normal saline on the pressure-volume loop in a healthy patient?

A. ○ Loop narrows because of an increase in end-systolic volume
B. ○ Loop narrows because of an increase in end-systolic volume and becomes taller because of an increase in end-systolic pressure
C. ○ Loop widens because of a drop in end-systolic volume
D. ● Loop widens because of an increase in end-diastolic volume
E. ○ Loop width stays the same, but end-systolic pressure increases

Hide Explanation

The correct answer is the loop widens because of an increase in end-diastolic volume (D). Infusing normal saline will increase the amount of blood returning to the heart, thus increasing the end-diastolic volume and widening the loop. The end-systolic volume stays the same, since the increase in preload will stimulate a greater force of contraction and empty the heart to the same end-systolic volume. Pressures remain the same. Loop narrowing due to an increase in end-systolic volume is seen in heart failure, where a drop in contractility reduces how much the heart can empty. A loop that narrows because of an increase in end-systolic volume (A) and becomes taller because of an i in cases of increased afterload, such as hypertension, which inhibits the heart's ability to empty and increases the end-systolic volume. Loop widening because of a drop in end-systolic volume (C) might be seen when an inotropic drug, like dobutamine or digitalis, increases the heart's force of contraction (contractility). A loop whose width stays the same but whose end-systolic pressure increases (E) would not be likely, because an increase in end-systolic pressure increases the afterload; the increase in afterload would decrease the heart's force of contraction and therefore the stroke volume.

4. Which of the following statements best describes the isovolumetric contraction phase on the pressure-volume loop?

A. ● Both the mitral and aortic valves are closed, and pressure increases
B. ○ Blood is ejected into the aorta, reducing ventricular volume
C. ○ Mitral valve is open, blood enters the left ventricle
D. ○ Rapid decrease in left ventricular pressure, but no change in volume

Hide Explanation

The correct answer is (A). Isovolumetric contraction occurs after the mitral valve closes but before the aortic valve opens. Since the ventricle contracts against a closed system, pressure rises without a change in volume. (B) describes ventricular ejection, (C) describes ventricular filling, and (D) describes isovolumetric relaxation, which occurs after systole.

5. A patient with chronic hypertension is expected to have what change on the pressure-volume loop?

A. ● Decreased stroke volume due to increased afterload
B. ○ Increased stroke volume due to increased preload
C. ○ Decreased end-systolic pressure due to reduced afterload
D. ○ Increased contractility due to reduced preload

Hide Explanation

The correct answer is (A). Chronic hypertension increases afterload, meaning the left ventricle must generate greater pressures and work harder to eject blood into the aorta. With more effort to generate pressures, there is less "energy" to eject blood. This results in more blood remaining in the ventricle at the end of systole, or a higher end-systolic volume (ESV) and thus lower stroke volume (SV = EDV- ESV). (B) is incorrect because increased preload can increase stroke volume, but hypertension primarily affects afterload. End-systolic pressure increases with hypertension, not decreases (C). Contractility is not directly increased by hypertension (D).

How would you rate this Brick?

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Review Learning Objectives

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Analyze and interpret a ventricular pressure-volume loop.~~
02 ☑ ~~Relate the ventricular pressure-volume loop to the events of the cardiac cycle.~~
03 ☑ ~~Determine stroke volume, ejection fraction, and stroke work from a pressure-volume loop.~~
04 ☑ ~~Describe how pressure-volume loops are affected by preload, afterload, and autonomic nervous system activity.~~

Objective Confidence: 4/4